Agent Architecture & Interaction Models

1. The environment an agent perceives

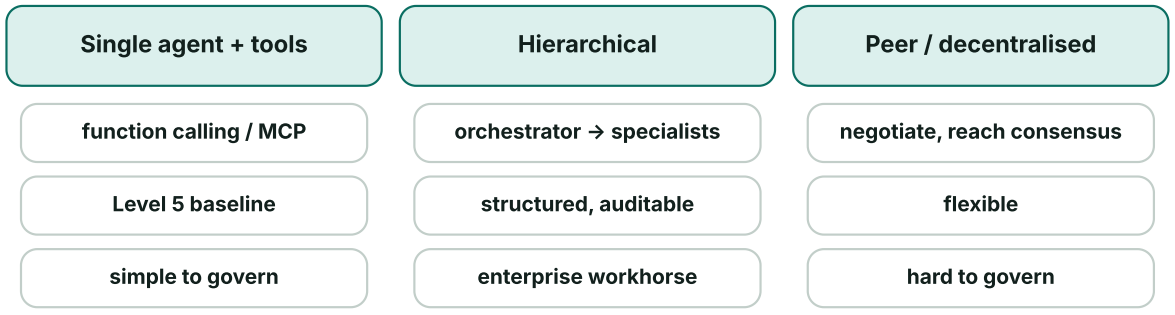
- **Digital context**: unstructured data (text, images), structured data (databases, **knowledge graphs**), vector stores for similarity search
- **Physical context**: sensors and actuators for embodied agents
- **Shared memory / communication hub**: in multi-agent systems, where agents exchange information, plans and goals

2. Memory is not one thing

- **Short-term (working) memory**: the current task and trace
- **Long-term memory**: past experience and learned facts
- **Shared memory**: team state
- **Knowledge graphs** give a structured understanding of entities and relationships that plain vector search can't, enabling deeper reasoning

3. Interaction models

✎ DIAGRAM



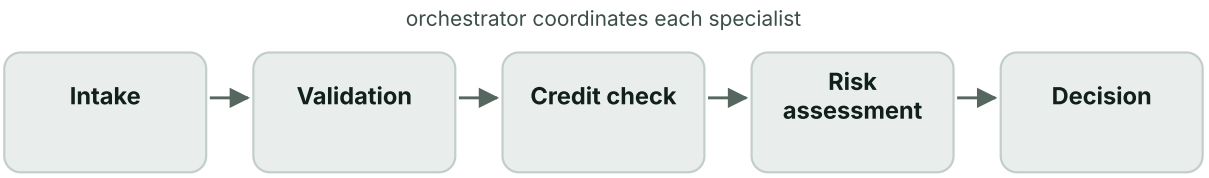
From simple to collaborative.

4. Two case studies

- **Travel planning agent**: one agent takes a goal ('a Tokyo trip within budget'), senses via flight and hotel APIs, plans an itinerary, books it. A clean single-agent example of the full loop
- **Agentic loan processing**: intake → validation → credit check → risk assessment → decision. As one agent it's a brittle monolith; as a multi-agent system an orchestrator coordinates specialists via A2A, which is more robust, maintainable and scalable

- Loan processing recurs because it has everything: multi-step flow, high stakes (governance, audit), clear specialisation, and exception paths

✎ DIAGRAM



The loan pipeline: each stage is a natural specialist agent.

SAY IT

"Hierarchical orchestration is the enterprise default because it's structured and auditable; I'd only go decentralised when the problem genuinely rewards it."